PART 1

Traffic, Delivery & Runtime Basics

1. Rate Limiting
2. Caching
3. Load Balancing
4. Reverse Proxies
5. API Gateways
6. CI/CD
7. Docker
8. Kubernetes

#001

Rate Limiting

WHAT IT IS

Restricting how many requests a client can make in a time window.

WHY IT MATTERS

Protects against abuse, brute force, and noisy neighbours; keeps the system fair and stable.

EXAMPLE

```php
// app/Providers/AppServiceProvider.php
RateLimiter::for('api', fn (Request $r) =>
    Limit::perMinute(60)->by($r->user()?->id ?:
    $r->ip())
);

// routes/api.php
Route::middleware('throttle:api')->group(function ()
    {
    Route::get('/tasks', [TaskController::class,
    'index']);
});
```

#002

Caching

WHAT IT IS

Storing the result of expensive work (queries, API calls, renders) so it can be reused.

WHY IT MATTERS

Cuts latency and database load dramatically.

EXAMPLE

```
$stats = Cache::remember("project:{$project->id}:stat
    s", now()->addMinutes(5), function () use (
    $project) {
    return $project->tasks()->selectRaw('status,
    count(*) c')->groupBy('status')->pluck('c',
    'status');
});
```

#003

Load Balancing

WHAT IT IS

Spreading incoming traffic across several identical app instances.

WHY IT MATTERS

Higher capacity and availability; one instance dying does not take you down.

EXAMPLE

```
upstream app {
    least_conn;
    server 10.0.0.11:8080;
    server 10.0.0.12:8080;
    server 10.0.0.13:8080 backup;
}
server { location / { proxy_pass http://app; } }
```

#004

Reverse Proxies

WHAT IT IS

A server that sits in front of your app and forwards requests (nginx, Caddy, HAProxy, Traefik).

WHY IT MATTERS

TLS termination, compression, static files, buffering slow clients, hiding internals.

EXAMPLE

```
// bootstrap/app.php
->withMiddleware(fn (Middleware $m) =>
    $m->trustProxies(at: ['10.0.0.0/8']))
```

#005

API Gateways

WHAT IT IS

A single entry point for many services that handles auth, routing, rate limits, logging and transformation.

WHY IT MATTERS

Clients talk to one URL; cross-cutting concerns live in one place.

EXAMPLE

```
# Kong declarative config (simplified)
services:
  - name: billing
    url: http://billing:8000
    routes: [{ name: billing-route, paths:
    [/billing] }]
    plugins: [{ name: rate-limiting, config: {
    minute: 100 } }]
```

#006

CI/CD

WHAT IT IS

Continuous Integration (auto build and test every push) and Continuous Delivery/Deployment (auto ship passing builds).

WHY IT MATTERS

Small, frequent, safe releases; regressions are caught before users see them.

EXAMPLE

```
name: ci
on: [push, pull_request]
jobs:
  test:
    runs-on: ubuntu-latest
    services:
      postgres: { image: postgres:16, env: {
    POSTGRES_PASSWORD: secret }, ports: ['5432:5432'] }
    steps:
      - uses: actions/checkout@v4
      - uses: shivammathur/setup-php@v2
        with: { php-version: '8.3' }
…
```

#007

Docker

WHAT IT IS

Packages your app and its dependencies into an image that runs identically anywhere.

WHY IT MATTERS

"Works on my machine" disappears; deploys become reproducible.

EXAMPLE

```
FROM php:8.3-fpm-alpine
RUN docker-php-ext-install pdo_pgsql opcache
WORKDIR /var/www
COPY --from=composer:2 /usr/bin/composer
    /usr/bin/composer
COPY . .
RUN composer install --no-dev --optimize-autoloader
CMD ["php-fpm"]
```

#008

Kubernetes

WHAT IT IS

An orchestrator that runs containers across a cluster, restarting, scaling and routing for you.

WHY IT MATTERS

Self-healing, declarative deployments, autoscaling, rolling updates.

EXAMPLE

```
apiVersion: apps/v1
kind: Deployment
metadata: { name: taskflow-web }
spec:
  replicas: 3
  selector: { matchLabels: { app: taskflow-web } }
  template:
    metadata: { labels: { app: taskflow-web } }
    spec:
      containers:
        - name: app
          image: registry.example.com/taskflow:1.4.2
…
```

Found this useful?

Save it for later.

Share it with a teammate.

Full guide with all code examples:
the Backend Engineering Handbook (GitHub).